

Edge4AV: Communication Latency for Connected and Automated Vehicle Applications

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Abstract

Are today's vehicle-to-everything (V2X) and fifth-generation (5G) communication paths ready for tomorrow's connected and automated vehicle (CAV) applications? Conventional connected-vehicle (CV) and intelligent transportation system (ITS) applications exchange compact, stateless SAE International J2735 updates. Complex CAV applications also require correlated, stateful operations that carry larger objects through event-triggered bursts or sustained streams in either direction. Prior field studies generally evaluate selected applications or one communication path. Consequently, they do not establish the workload and operating conditions under which both classes meet application deadlines across direct and network-assisted communication.

We conduct a controlled real-vehicle study of application communication over a commercial Long-Term Evolution cellular V2X (LTE C-V2X) direct PC5 path and a private 5G New Radio (NR) network-assisted Uu path. The Intersection Programming Interface (IPI), an enabling application-layer protocol, supplies a compact profile for typed or pre-encoded J2735 messages and a correlated-operation profile for CAV workloads. The 5G deployment uses one physical CAV user equipment (UE), a dedicated radio, and a clean 40-MHz n48 channel without ambient contention. We vary payload size, direction, transport, radio condition, time-division-duplex configuration, competing traffic, concurrent demand, and interruption. The metrics are response round-trip time (RTT), the percentage of issued requests answered within each application deadline, and directional application goodput.

The measurements yield three insights. 1) The measured commercial direct V2X path supports compact J2735 messages only within a limited payload and coverage envelope. 2) The measured 5G uplink supports only severely limited CAV packet sizes within decision deadlines. 3) 5G and sixth-generation (6G) radios must support event-triggered bursts and sustained streams in either direction while isolating concurrent flows. These capabilities are necessary for CAV and ITS applications that advance the goal of zero road fatalities.

1 Introduction

Connected vehicles (CVs), including human-driven vehicles, and connected and automated vehicles (CAVs) need information that may be visible only to other vehicles or roadside infrastructure. An occluded road user, the current traffic-signal state, or a cooperative maneuver plan can change

a driving decision. Vehicle-to-everything (V2X) and fifth-generation (5G) communication are therefore expected to deliver this information to automated driving systems and human drivers [2, 28].

The applications create two communication patterns. CV and intelligent transportation system (ITS) applications commonly exchange compact, stateless SAE International J2735 updates. A new update can replace an older one without retaining an application session. Complex CAV applications also require correlated, stateful operations for cooperative perception, trajectory planning, maneuver coordination, and remote assistance. An event or fault can trigger an uplink-heavy or downlink-heavy burst, whereas remote teleoperation can sustain video and control traffic. These workload classes differ in object size, direction, duration, and deadline, but both are proposed for direct V2X and network-assisted 5G communication.

Existing standards and systems provide the radio paths, message formats, and service procedures needed by parts of this vision. Existing measurements also establish important performance results for individual applications and paths. However, the literature we examine does not compare the workload and operating conditions under which compact CV/ITS messages and correlated CAV operations meet their application requirements over both path classes. We therefore ask: *Can today's V2X and 5G paths support both CV and CAV applications? At what workload size and operating condition does each path stop meeting the required response latency, completion rate, or application goodput?*

We answer this question with a controlled, real-vehicle field study. The study applies both workload classes to a commercial Long-Term Evolution cellular V2X (LTE C-V2X) direct PC5 path and a private 5G New Radio (NR) network-assisted Uu path. To represent the workloads consistently, we develop the Intersection Programming Interface (IPI). IPI is an application-layer protocol that preserves typed or pre-encoded J2735 messages and represents correlated CAV operations. The private network contains one physical CAV user equipment (UE), a dedicated radio, and a clean 40-MHz n48 channel; all competing traffic is introduced by the experiments. Across payload size, direction, transport, radio condition, configuration, competing traffic, application demand, and interruption, we measure response round-trip time (RTT), the percentage of issued requests answered

within each application deadline, and directional application goodput.

This paper makes two contributions:

- **A controlled real-vehicle measurement study.** We evaluate representative CV/ITS messages and correlated CAV workloads over direct PC5 and network-assisted Uu. The experiments identify when payload, direction, radio condition, configuration, and shared demand cause each measured path to miss an application deadline or required directional rate.
- **An enabling CV/CAV application protocol.** IPI provides one application contract for typed or pre-encoded J2735 messages and correlated CAV operations. Distinct message and operation profiles preserve message content and operation association, while path-specific bindings carry the resulting frames over the measured PC5 and Uu paths.

The measurements yield three insights:

- 1) **The measured commercial direct V2X path supports compact J2735 messages only within a limited payload and coverage envelope.** Roadside-unit placement and supplemental coverage must account for buildings, highway geometry, and sparse infrastructure.
- 2) **The measured 5G uplink supports only severely limited CAV packet sizes within decision deadlines.** CAV developers must reduce or progressively transmit vehicle-originated state when a full object cannot meet its deadline.
- 3) **5G and sixth-generation (6G) systems must support event-triggered CAV bursts, sustained directional streams, and concurrent deadline-sensitive traffic.** Radio configurations and schedulers must preserve both directions and isolate simultaneous flows by deadline and duration.

2 Related Work

Answering the readiness question requires comparable field evidence for both workload classes. Prior work provides interoperable messages, network and edge procedures, systems for individual CAV applications, and measurements of direct or network-assisted paths. Figure 1 organizes these foundations around the capabilities required for cooperative transportation. The primary gap is a common application-level evaluation of compact CV/ITS updates and correlated CAV operations over both path classes. Such an evaluation also needs one application contract that represents the two workload classes consistently.

Standards and application platforms. Vehicular interoperability is divided among application messages, network services, and edge information exchange. SAE International

J2735 defines safety, road-user, map, signal, and priority messages for CV and ITS applications [30]. The Third Generation Partnership Project (3GPP) defines PC5 and Uu communication and a V2X Application Enabler with registration, delivery, resource-management, continuity, and session procedures [1, 3]. The European Telecommunications Standards Institute (ETSI) Multi-access Edge Computing (MEC) V2X Information Service adds PC5/Uu provisioning, predicted quality of service, publication, and subscription procedures [15]. Architectures and platforms from the 5G Automotive Association, ETSI, V2X Hub, and CARMA connect selected actors and deployment functions [5, 14, 35, 36].

Together, these mechanisms define complementary forms of interoperability. J2735 defines individual application messages, 3GPP coordinates V2X application services and sessions, and MEC distributes V2X information. We found no surveyed implementation that combines byte-preserving J2735 carriage with generic, correlated CAV operations behind one application contract for both PC5 and Uu. The evaluation needs this combination to preserve workload meaning and operation association as the communication path changes. IPI supplies the enabling contract without replacing the message, service, or network standards beneath it.

CAV systems and middleware. Networked CAV systems demonstrate why application representation affects communication demand. EMP and AutoCast select cooperative-perception information [29, 41], while VIPS, VI-Map, and SOAR return fused perception, compact maps, or coordinated infrastructure results [19, 32, 33]. Their transmitted objects include intermediate features, detections, trajectories, and control state. The chosen representation determines the transmitted size, and the driving task determines the deadline.

Application-aware middleware then adapts execution to these requirements. D3 schedules autonomous-driving operators against end-to-end deadlines [18]. Tentacles selects or combines networks, adapts data quality, and invokes local fallback when a remote result misses its deadline [39]. AutoWareV2X selects the fresher collective-perception message received over Wi-Fi or LTE [7]. These systems show how application state can guide network selection and fallback, but each system defines the exchanged object and operation state for its own application. IPI makes message type, operation identity, timing, and generic outcome available through one application contract. This information can support future network-selection and adaptation mechanisms for both workload classes.

Direct V2X and cellular field evidence. Direct-path studies connect radio behavior to complete application objects. SEE-V2X combines commercial C-V2X traces with awareness and perception workloads. Ku et al. measure fragmented-image delivery, National Institute of Standards

Toward Zero-Fatality Cooperative Transportation

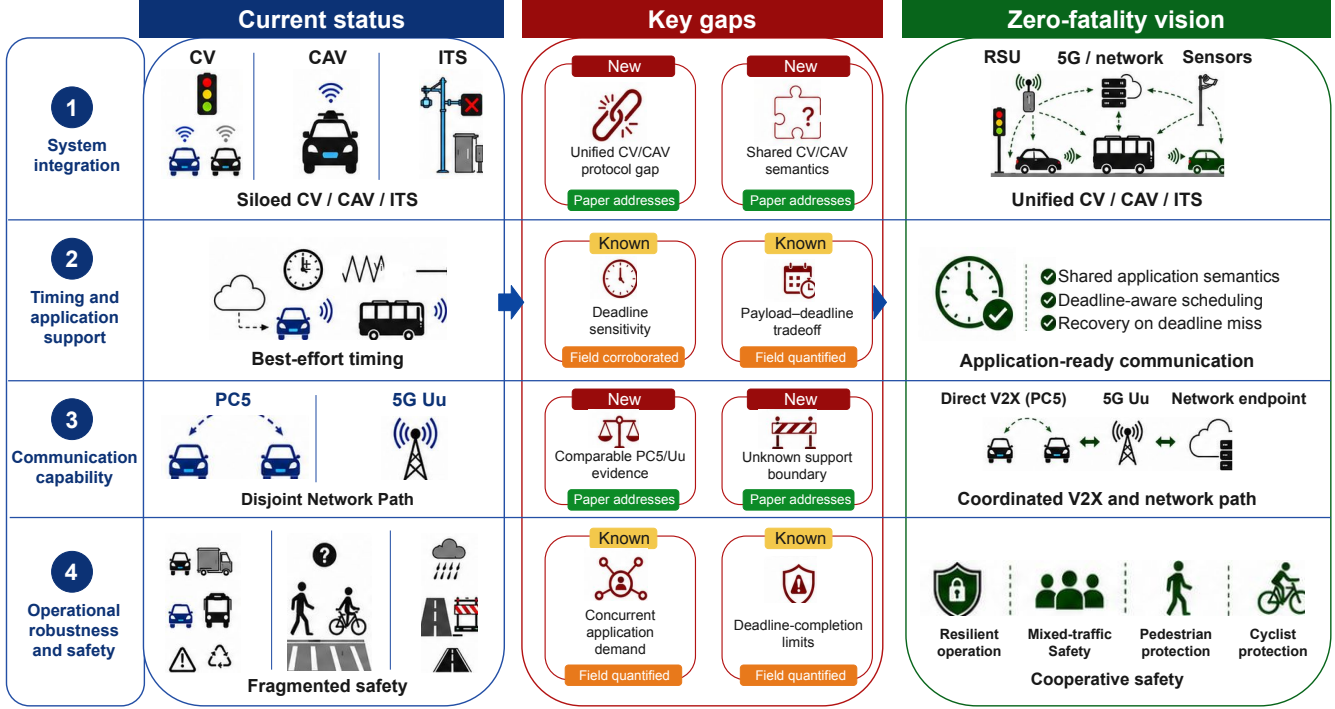


Figure 1: Communication capabilities required for cooperative transportation. Existing technologies address individual message, application, and path functions. The measurement study determines when the two workload classes meet their application requirements, and IPI supplies the application contract used to represent them.

and Technology testing examines device interoperability, and recent hardware-in-the-loop and CooperScene studies connect PC5 behavior to control or perception [24, 25, 34, 38, 40]. Collectively, these studies show that packet reception alone can overstate application support because one missing fragment or late update can invalidate an object.

Cellular studies reach the same conclusion through metrics selected for each application. Vehicle-upload, teleoperation, and TARGET-X trials report throughput, frame latency, freshness, service quality, or edge behavior [9, 17, 23, 26]. The Next Generation Mobile Networks Alliance identifies implementation and interoperability requirements for balanced and uplink-oriented time-division-duplex frames [27]. Demircioglu’s 40-km private 5G standalone corridor provides the closest cross-path baseline: it integrates PC5 and Uu and measures application-layer latency and packet delivery for six Cooperative ITS scenarios across speed and traffic-density conditions [13].

This literature establishes deployable paths, representative applications, and important path-specific measurements. Its studies use different application objects, denominators, and timing criteria. They therefore do not determine the

conditions under which both workload classes meet their deadlines across direct PC5 and network-assisted Uu. The real-vehicle study fills this measurement gap by retaining every issued request and testing each response against an application deadline across payload, direction, radio condition, continuous traffic, concurrent demand, and interruption. Section 3 next defines the IPI representation that makes these comparisons possible. Sections 4 and 5 then map the workloads to the two paths and measure their application limits.

3 IPI Protocol Design

A comparable evaluation of compact J2735 updates and correlated CAV operations requires a workload representation that preserves existing J2735 content and associates the requests, intermediate updates, and outcomes that belong to one CAV operation. The Intersection Programming Interface (IPI) provides this application-layer contract. It enables the measurement study and can also support applications that use both workload classes.

Application interface, wire protocol, runtime, and bindings. IPI comprises four components. First, the application programming interface lets an application publish or

receive a message, invoke an operation, retrieve its responses, and manage an application session. Second, a transport-independent wire protocol defines separate serialized profiles for messages and operations. Third, the reference runtime validates frames and maintains local session and response records. Fourth, a binding places each serialized frame in PC5 custom data or an Internet Protocol transport on Uu. Commonality resides in the application contract and operation semantics; each profile serializes only the fields required by its workload and binding.

Compact messages and correlated operations. The compact message profile carries an individual J2735 message. Typed helpers cover Basic Safety Message (BSM), Personal Safety Message (PSM), Map Data (MAP), Signal Phase and Timing (SPaT), Signal Request Message (SRM), and Signal Status Message (SSM). The byte-preserving form wraps a complete pre-encoded J2735 message with a one-byte type and a four-byte length. It therefore retains the deployed message bytes without adding operation state.

The correlated-operation profile represents perception, planning, control, and map interactions. A request starts an operation. An update carries nonterminal progress or intermediate content. A completion carries a successful terminal result, and a rejection reports refusal or unsuccessful termination. The correlation identifier associates each update or terminal result with its request. The session identifier groups related operations within a continuing interaction. Expiration identifies when an operation or result is no longer useful. Registration establishes the application session, heartbeat refreshes its lease, telemetry reports session-associated state, and termination closes the session. An IPI application session is distinct from a Transmission Control Protocol (TCP) connection, a Message Queuing Telemetry Transport (MQTT) session, and a 3GPP protocol data unit or V2X Application Enabler session.

Typed planning, perception, and control content supports known objects. Opaque content carries an application object together with its declared service class when both endpoints use an external schema. Thus, the operation profile adds the association and outcome fields required by stateful CAV interactions while the compact profile leaves J2735 messages small.

Responsibility boundary. IPI standardizes application representation, operation association, generic outcome, and freshness. A binding supplies frame delimitation, datagram formation, fragmentation and reassembly, delivery, retransmission, and congestion behavior. The PC5 binding places one complete IPI frame in one vendor custom-data payload. The TCP binding uses a four-byte network-order length prefix. The MQTT binding places one serialized frame in one MQTT application payload and uses Quality of Service (QoS) 0. Raw User Datagram Protocol (UDP) places one frame

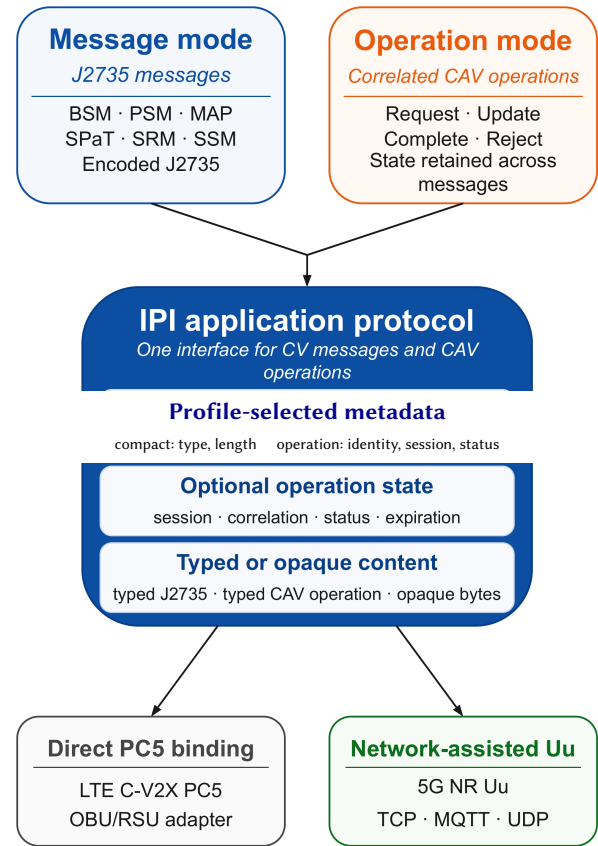


Figure 2: IPI presents one application contract for compact J2735 messages and correlated CAV operations. The compact and operation profiles use distinct serialized fields, and the selected binding carries the resulting frame over direct PC5 or network-assisted Uu.

in one datagram when it fits; the 1,400-B fragmentation and reassembly evaluated later belong to the UDP measurement adapter. The deployment configures security, admission, radio scheduling, and path selection. The application determines domain-specific decisions such as whether to accept a maneuver. These responsibilities preserve application meaning while allowing communication behavior to change across the paths evaluated in this paper.

Figure 2 shows the resulting organization. The application contract presents message and operation modes. Their serialized profiles retain the compact or correlated fields needed by the selected PC5 or Uu binding.

Reference implementation. The C++17 implementation provides the four components above. Its tests exercise typed and pre-encoded J2735 messages, independently generated J2735 IPI conformance vectors, correlated operation roles,

Table 1: IPI requirements, mechanisms, and supporting evidence.

Requirement	IPI mechanism	Evidence in this paper
Carry deployed CV/ITS messages	Typed helpers and byte-exact pre-encoded J2735 content	Callback, round-trip, and byte-preservation tests
Represent a CAV operation lifecycle	Request, update, completion, rejection, session, correlation, and expiration fields	Lifecycle, correlation, and malformed-frame tests
Use one application contract across paths	Distinct message/operation profiles and path-specific adapters behind one API	PC5 custom-data and Uu field campaigns
Control representation size	Five-byte compact wrapper and bounded operation fields	Measured byte counts in Section 5.1

session lifecycle, malformed input rejection, PC5 framing, and broker-backed operation exchange. Section 5.1 measures the representation cost and workload coverage. Section 4 next identifies the serialized IPI profile and payload-size boundary used by each field campaign.

4 System Design and Experimental Setup

The evaluation uses IPI to represent compact CV/ITS messages and correlated CAV operations before mapping them to the workload sizes and directions tested over direct PC5 and network-assisted Uu. We measure response RTT and deadline completion for each application workload. Separate exact directional transfers measure receiver-observed application goodput after byte-count and checksum validation. The study covers the two access modes emphasized by the 3GPP V2X architecture: direct PC5 communication and network-assisted Uu communication [1]. PC5 exchanges data locally between vehicles and roadside infrastructure. Uu carries vehicle traffic through a radio access network and mobile core to an application endpoint. The experiments use one commercial implementation of each path on the same autonomous vehicle. Protocol fields and serialization additions are reported in bytes (B). Application payloads and transferred objects use kibibytes (KiB = 2^{10} B) or mebibytes (MiB = 2^{20} B). Link rates and application goodput use decimal kilobits per second (kbit/s), megabits per second (Mbit/s), or gigabits per second (Gbit/s).

4.1 Testbed and Measured Paths

IPI defines the application message and request/response correlation for both measured paths. A path-specific adapter then binds the serialized IPI exchange to PC5 or Uu. Figure 3 shows the corresponding physical installation and radio deployment. The vehicle host generates each IPI workload and records sender-side RTT. A Mocar on-board unit (OBU) provides the PC5 interface, while an MG52 or GL-X3000 gateway provides the Uu interface. At the fixed site, a HUALI roadside unit (RSU) terminates PC5, and an outdoor Airspan AirSpeed 2900 gNodeB terminates Uu. Both paths reach the d1 application host through direct wired connections to the RSU or the

on-site MX250. The remote MX68CW manages the private 5G deployment and remains outside the measured user-data path [10–12].

In the July 3–4 PC5 campaigns, the PC5 adapter bound IPI to the vendor’s 2020 J2735 stack. We first exercised the standard-message callbacks to validate the J2735 path. Performance probes used the serialized IPI profile IPI_RTT1. The adapter placed the complete frame in the application payload of the vendor’s custom packet-data interface, packet type $0 \times 1b$. The frame records the message kind, origin and responder identifiers, sequence number, endpoint timestamps, and configured frame length. The RSU decodes these fields and returns a response that copies the origin identifier, sequence number, initial send time, and configured length. Thus, the sequence field inside the IPI performance frame correlates each response with its request. For the 256–2,048-B conditions, the request and response each use the configured complete frame length. The zero setting produces the minimum valid header rather than a zero-length packet. The installed interface’s 4,080-B limit applies to the complete serialized packet submitted to the vendor interface. The vendor reports protocol-conformance certification for the product families [21].

The Uu measurement adapters carry serialized IPI operation frames over TCP, MQTT, or UDP. Most campaigns connect the vehicle host to a Cisco Meraki MG52 gateway over 1-Gbit/s Ethernet. The MG52 contains the subscriber identity module (SIM) and an internal antenna placed face-up on the trunk floor. The August 19 directional repeat replaces the MG52 with a GL.iNet GL-X3000 gateway and evaluates both time-division duplex (TDD) configurations with that device. Each gateway reaches d1 through the AirSpeed gNodeB, the Cisco private 5G system, and the on-site MX250. Both measured paths therefore use IPI at the application layer. PC5 uses the vendor custom-data binding, whereas the Uu workload adapters use TCP, MQTT, or UDP.

Figure 4 expands these components into the measured IPI request and response paths. The PC5 binding carries a serialized IPI performance request and returns a response with the same configured frame size for every nonzero condition. The

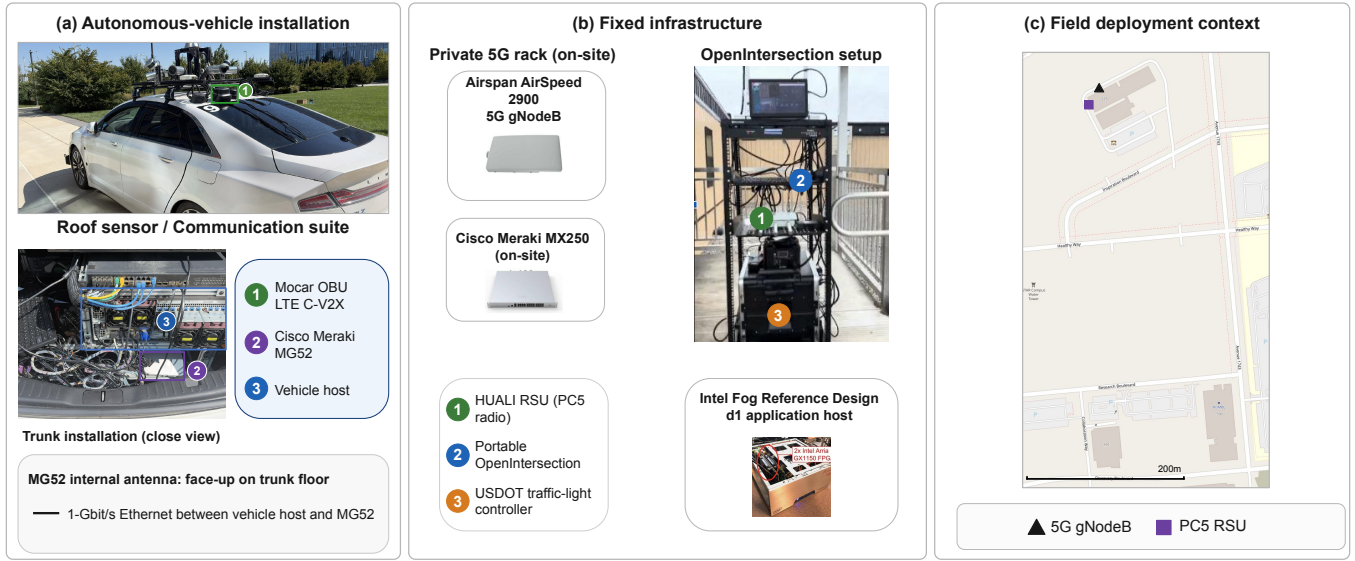


Figure 3: Physical testbed. (a) The autonomous vehicle carries the Mocar OBU, MG52 gateway, and vehicle host. (b) The fixed installation contains the AirSpeed 2900 gNodeB, MX250, HUALI RSU, traffic-light controller, portable OpenIntersection, and d1 application host. (c) The map locates the gNodeB and PC5 RSU. Map data © OpenStreetMap contributors.

response copies the request’s origin and sequence values. A Uu request carries an IPI Cooperative Service frame. In an uplink-heavy exchange, the vehicle sends the declared application object, and d1 returns a compact correlated acknowledgment after decoding it. In a downlink-heavy exchange, the vehicle sends a compact request, and d1 returns the requested application object. The vehicle validates the response sequence number, length, and checksum before stopping its RTT timer.

The Uu deployment was tightly controlled. The vehicle gateway was the sole UE, and the AirSpeed radio and its full 40-MHz channel were dedicated to the experiment. The Citizens Broadband Radio Service (CBRS) n48 band was clean and carried no ambient traffic. The baseline measurements contained no unrelated user devices or ambient radio-band contention. Background streams and 1–100 application clients were introduced only as controlled experimental variables behind the same UE attachment. Table 2 summarizes the controls that bound the application measurements and identifies the comparison enabled by each control.

The 5G radio uses 30-kHz subcarrier spacing and TDD. The operator reported 70/20/10 as the deployment’s established allocation. These values denote the downlink, uplink, and dynamic shares. All application campaigns before the TDD study used this profile. Its downlink-oriented allocation is consistent with the downlink-heavy frame structures reported for current 5G deployments [27]. We also evaluate 40/40/20, which assigns 40% of the frame to fixed uplink

use rather than 20%. Both profiles retain 10D4G frame packing. The TDD study includes repeated uplink-heavy RTT sweeps, matched downlink-heavy RTT sweeps, and exact 50-MiB transfers in each direction. A replacement-gateway repeat uses the GL-X3000 for complete uplink-heavy and downlink-heavy matrices and three exact transfers per direction under each profile. Each run records the application direction, location, and available signal context; the earlier runs additionally record available cell state and serving-cell context. The Airspan distributed-unit (DU) Cell exports provide five-minute cell volume, active-time, and unavailability counters for the available windows. These counters verify the serving cell and measurement-block context. Endpoint records define application RTT and goodput.

A Samsung 22 handset running G-NetTrack Pro recorded radio conditions during three continuous drives through the testbed. The combined trace contains 1,301 Global Navigation Satellite System (GNSS) records. We remove 123 records from a frozen radio-value interval longer than 5 s and retain 1,178 valid NR band-48 records. Reference Signal Received Power (RSRP) spans -121 to -88 dBm, with a median of -108 dBm. The corresponding NR signal-to-interference-plus-noise ratio (SINR) spans -20 to 30 dB, with a median of 9 dB. G-NetTrack Pro’s ordinary signal-to-noise ratio field is empty. We recover SINR from the timestamped NR `ssSinr` entries in the verbose logs and align each entry to its nearest GNSS record.

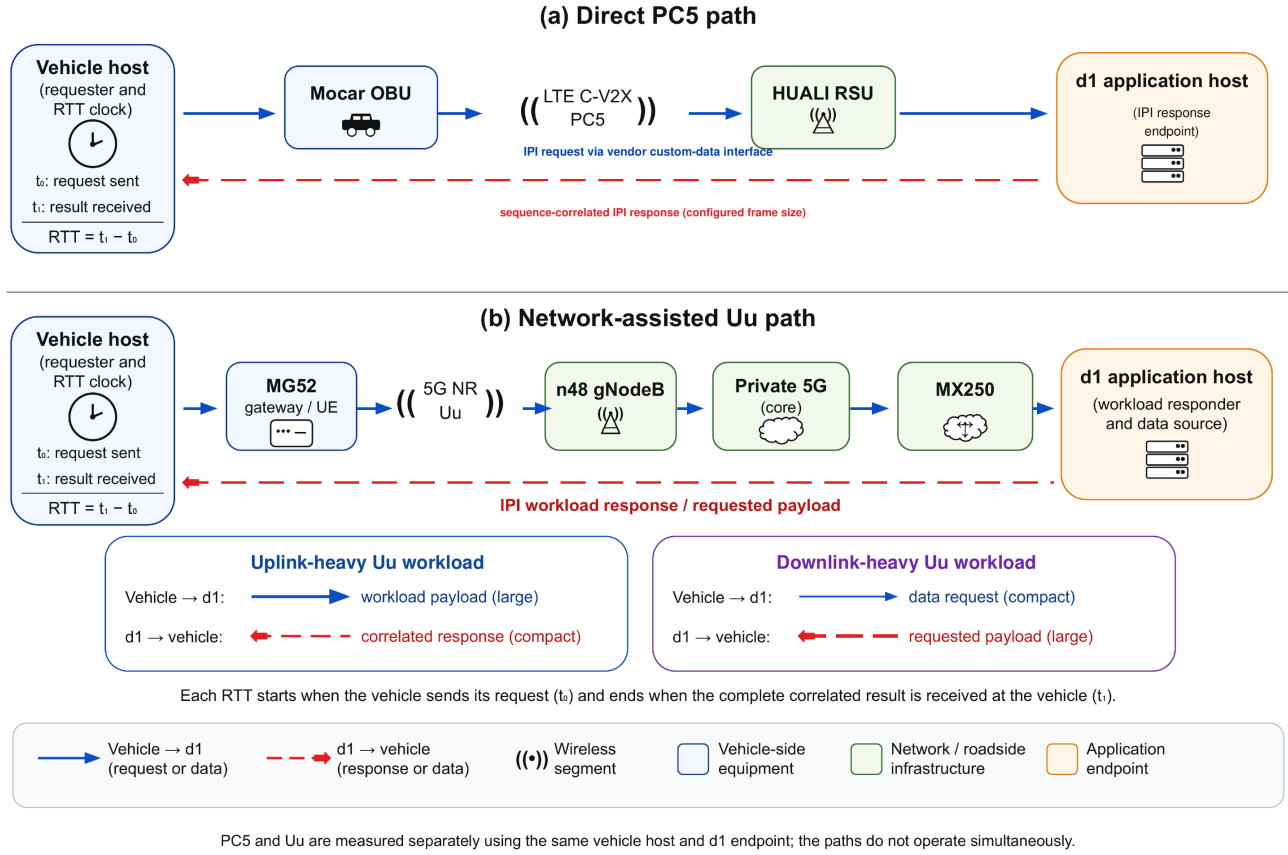


Figure 4: IPI path bindings and sender-side RTT boundaries. (a) The direct PC5 adapter carries a serialized IPI performance request through the vendor packet-data interface and returns a sequence-correlated IPI response with the same configured frame size for every nonzero condition. (b) The network-assisted Uu adapter carries either a large vehicle-originated IPI workload followed by a compact correlated response or a compact IPI request followed by the complete requested payload. PC5 and Uu were measured separately.

Table 2: Experimental controls and their roles in the analysis.

Dimension	Controlled condition	Role in comparison
Radio occupancy	Dedicated 40-MHz n48 channel; no ambient users or traffic	Establishes the single-UE baseline; load is introduced only by the experiment
Client population	One physical CAV UE; 1–100 logical clients share its attachment	Varies application concurrency without changing the number of radio clients
Endpoint and timing	d1 directly connected to the RSU and MX250; unsynchronized endpoint clocks	Supports sender-side RTT and validated goodput; synchronized one-way delay is unavailable
Field context	Stationary workload records; continuous-drive radio survey; location-joined PC5 outcomes	Relates application measurements to stationary conditions and route coverage
TDD and gateway	70/20/10 and 40/40/20; MG52 and GL-X3000 repeats	Tests whether configuration behavior persists across vehicle gateways and both traffic directions

3GPP defines the 5G Synchronization Signal Reference Signal Received Power (SS-RSRP) measurement but does not

assign qualitative coverage levels [4]. We aggregate the outdoor pedestrian levels in Australia’s 2026 coverage-map standard [8]: Good maps to strong ($\text{RSRP} \geq -95$ dBm), Moderate

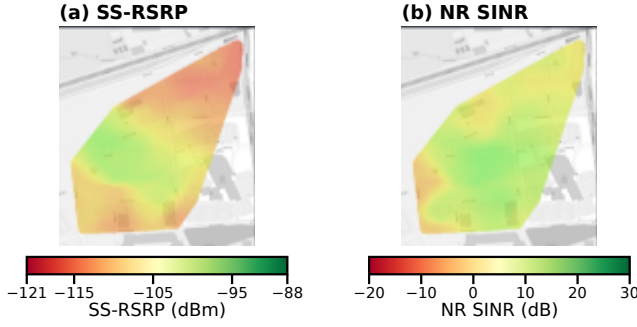


Figure 5: Continuous-drive radio survey. A Samsung 22 handset running G-NetTrack Pro recorded 1,178 valid n48 samples across three drives. Panels interpolate (a) SS-RSRP and (b) NR SINR within the sampled boundary after removing frozen radio values longer than 5 s. Map data © OpenStreetMap contributors.

maps to common/typical ($-105 \leq \text{RSRP} < -95$ dBm), and Basic and No Coverage map to weak ($\text{RSRP} < -105$ dBm).

The temporal sample distribution contains 83 (7.0%) strong, 340 (28.9%) common/typical, and 755 (64.1%) weak records. We use the drive survey to characterize route-level coverage and each workload campaign’s signal record to compare stationary application measurements. Figure 5 maps local interpolation of RSRP and SINR within the sampled boundary. The PC5 route analysis joins every issued request and received response to GNSS coordinates, producing spatial response-completion and RTT measurements along each driven path.

Figure 4 defines the request and response events at the RTT boundary. The PC5 RTT covers the IPI-to-PC5 adapter, vendor packet-data interface, two direct-radio traversals, response handling, and correlation. The Uu RTT covers the IPI transport adapter, vehicle host, gateway, radio access, private core, local routing, d1 responder, and return path. The d1 responder decodes each workload frame and immediately returns its response. Its direct connections to the RSU and MX250 make local wired transfer negligible relative to the measured communication paths. The clocks were not synchronized, so all timing uses the sender’s round-trip clock. Each timer begins immediately before the vehicle transmits its request. An uplink-heavy timer ends when the vehicle receives the compact acknowledgment. A downlink-heavy timer ends after the vehicle receives and validates the complete requested payload. The exact 50-MiB transfer experiment uses a separate endpoint timer. The receiver verifies the payload size and checksum before reporting the transfer duration and application goodput.

4.2 Application Conditions and Workloads

Connected-vehicle and CAV applications place different data sizes, dominant directions, durations, and deadlines on the communication path. Table 3 maps six representative applications to the workload probes used to evaluate these requirements [2, 6, 20, 37]. Periodic signal and collision warnings create compact shared updates. For emergency-vehicle signal priority, an authorized vehicle sends an SRM to request priority or preemption, and the infrastructure returns the request status in an SSM [31]. Remote assistance for a disabled or malfunctioning CAV can begin with a compact request and return perception, planning, control, or other data from an edge or help-center endpoint. Remote recovery can also add a continuous vehicle-originated video stream. At 25 mph, the vehicle moves 0.11 m in 10 ms and 1.12 m in 100 ms. Under Federal Highway Administration assumptions, its level-road stopping sight distance is about 46.3 m [16]. A 35-mph adjacent vehicle changes its gap to a 25-mph vehicle by 0.45 m in 100 ms. Remote assistance below 10 km/h moves a vehicle at most 0.278 m in 100 ms and 0.556 m in 200 ms. These applications cannot be represented by one payload, direction, duration, or deadline. Their traffic can be periodic, event triggered, or continuous. Perception, intent, and fault state create vehicle-to-edge demand. A compact vehicle request can instead return raw sensor data, processed perception, planning data, control data, or other state from the edge. Several operations can also share the same attachment while a continuing stream remains active.

The application conditions define four parts of the workload matrix:

- 1) **Payload and direction.** PC5 spans configured complete IPI frame sizes from the minimum-header control through 2 KiB. Uu spans 0–2 MiB using a vehicle-originated object followed by a compact acknowledgment. The downlink-heavy probe returns an exact 1-, 10-, 100-, or 500-KiB object after a compact vehicle request. Separate 50-MiB transfers measure validated goodput in each direction.
- 2) **Application-object size.** The Uu sweep includes 922 V2X-Radar object lists: 19.2 KiB at the median, 23.4 KiB at the 99th percentile, and 24.4 KiB at the maximum. Additional dataset objects span 8 KiB–2 MiB, and a 58.6-KiB object provides a stress condition.
- 3) **Transport.** Uu measurement adapters carry matched IPI workload frames through TCP, MQTT over TCP, raw UDP, and UDP with adapter-level 1,400-B fragments.
- 4) **Demand, configuration, and recovery.** The experiments add one to four sustained streams, 1–100 logical clients, a requested QoS label, two TDD profiles, two

Table 3: Representative application requirements and workload probes. $\uparrow$ denotes vehicle-to-edge traffic and $\downarrow$ denotes edge-to-vehicle traffic. Object sizes exclude transport and security bytes.

Application	Traffic pattern	Object size or rate	Deadline	Workload probe
Signal phase/status	Periodic infrastructure $\downarrow$	About 0.5 KiB at 10 Hz	100 ms	SPaT callback; compact PC5/Uu cycles
Collision or blind-spot warning	Periodic many-to-many PC5	About 0.2 KiB per source at 10 Hz	100 ms	0.25-KiB serialized IPI frames at stationary points and on routes
Emergency-vehicle signal priority	SRM $\uparrow$; SSM $\downarrow$	0.3–1 KiB	10 ms	SRM/SSM callbacks; 0.25–1-KiB cycles
Emergency maneuver	Reciprocal event-triggered burst $\uparrow\downarrow$	2–12 KiB	3–10 ms	2-KiB serialized IPI frames over PC5; 8–16-KiB Uu cycles
Remote-assistance data request	Compact request $\uparrow$; application object $\downarrow$	1–500 KiB	100–1,000 ms	Downlink-heavy TCP/MQTT cycles
Fault-triggered remote recovery	Video/state $\uparrow$; control/path $\downarrow$	Up to 1 KiB control; 32-Mbit/s video	100/200 ms	Bidirectional Uu; exact transfer, load, demand, and restart

vehicle gateways, and a 10-s endpoint or broker interruption.

These factors distinguish payload-size effects from transport, direction, configuration, competing traffic, demand, and interruption.

In the request/response experiments, each application client waits for its response or timeout before issuing the next request. At negligible response time, a 200-ms interval allows at most five requests per second per client. The 100 application clients remain application flows behind one vehicle gateway. The resulting matrix uses one application contract while comparing payload size, transport, direction, allocation, demand, and interruption through the same communication metrics. Table 4 orders the comparisons from application representation and single-path behavior to shared demand and interruption.

All request/response experiments use the same metrics. The endpoint clocks were not synchronized, so sender-side response RTT is the communication-latency metric. For deadline D , we define the deadline-completion rate as the fraction of issued requests answered within D :

$$C(D) = \frac{N_{\text{responses received within } D}}{N_{\text{issued requests}}}. \quad (1)$$

A PC5 timing cycle models a request/status exchange such as the SRM-to-SSM direction in Table 3. The vehicle sends a request, and the infrastructure returns a status message. The measured IPI performance response is matched through its origin and sequence fields. In an uplink-heavy Uu exchange, the response is a compact acknowledgment returned after the workload frame is decoded. In a downlink-heavy Uu exchange, the response is the complete requested payload, and its vehicle-side length, sequence, and checksum validation is included in RTT. Every issued request remains in the denominator; missing, rejected, timed-out, and late responses

reduce $C(D)$. RTT percentiles describe only the responses that arrived and are always reported with $C(D)$. We report the median (p50), 95th percentile (p95), and 99th percentile (p99) response RTT. We evaluate D at 3, 10, 25, 100, 200, 500, and 1,000 ms, matching the cooperative, warning, remote-assistance, and relaxed-transfer conditions above. For each application comparison, a measured condition crosses its timing boundary when response p95 exceeds the application deadline. The value $C(D)$ reports how many issued requests satisfy that deadline. The metrics provide different information: RTT characterizes received responses, and $C(D)$ counts every issued request against the deadline.

The exact bulk transfers use receiver-observed application goodput rather than $C(D)$. Each repetition verifies the transferred byte count and checksum, and goodput is the validated payload size divided by receiver-observed transfer duration. The MG52 blocks contain ten repetitions per profile and direction; the GL-X3000 blocks contain three. We report the individual transfers, mean, and a t -based 95% confidence interval. These intervals describe within-block transfer variation. Together, the workload mappings, timing boundaries, and issued-request denominators define the comparisons used in Section 5 to determine the workload conditions that each path supports.

5 Experiment Results

Each result uses the response RTT and deadline-completion definitions from Section 4 and the application conditions in Table 3. The analysis first establishes IPI workload coverage and direct-PC5 behavior. It then examines 5G uplink payloads, transport, and stationary radio conditions, followed by downlink-heavy workloads and two TDD configurations. The final experiments add sustained streams, concurrent clients, and interruption. This order separates single-path

Table 4: Experiment comparisons. $\uparrow$ denotes vehicle-to-edge traffic, and $\downarrow$ denotes edge-to-vehicle traffic.

Experiment	Varied condition	Measured outcome
IPI representation	J2735 message profile; correlated-operation profile	Added bytes; represented application content
Payload and deadline	PC5 minimum-header control and 0.25–2-KiB complete frames; Uu 0–2-MiB application objects; 3–1,000 ms	Completion; RTT boundary
Field and route coverage	Stationary points; four PC5 routes	Spatial completion; response tail
Transport	TCP; MQTT; raw UDP; UDP adapter fragmentation	Decode/reassembly completion; RTT
Direction	Object $\uparrow$ plus compact acknowledgment; request plus object $\downarrow$; 50-MiB $\uparrow/\downarrow$	Directional RTT; validated goodput
TDD and gateway	70/20/10 and 40/40/20; MG52 and GL-X3000	Stability; latency; goodput
Shared demand	Sustained streams; 1–100 clients; application label	Foreground completion; achieved rate
Interruption	10-s endpoint restarts	Missing attempts; recovery gap

Table 5: Measured IPI content representation for compact J2735 messages and correlated CAV planning operations.

IPI profile and workload	IPI fields (B)	Content (B)	Total (B)
CV message: preserved J2735	5	P	$P + 5$
CAV operation: planning request	63	0	63
CAV operation: request + object	78–81	256	334–337
CAV operation: request + object	78–81	1,024	1,102–1,105
CAV operation: request + object	78–81	4,096	4,174–4,177

limits from the conditions that place several CAV flows on the same network.

5.1 IPI Workload Representation and Framing Cost

Before measuring either communication path, we quantify how IPI represents the two workload classes and how many bytes the representation adds. The message profile preserves typed BSM, PSM, MAP, SPaT, SRM, and SSM content and a byte-exact pre-encoded Traveler Information Message (TIM). The operation profile represents requests, updates, completions, and rejections with session and correlation information.

Table 5 reports the IPI content representation before path-specific experiment and transport frames. A preserved J2735 payload of size P B receives a 1-B type field and a 4-B length field. The CAV rows contain a planning request and, when present, an opaque application object.

Each CAV object-size condition contains 1,000 encoded sender records. Their IPI fields encode the session and vehicle identifiers, service class and status, request horizon, planning waypoint, confidence, and task identifier. These fields occupy 78–81 B when a request carries an object. Their share of the encoded frame therefore decreases from 30.5–31.6% with a 256-B object to 7.6–7.9% with a 1-KiB object and 1.9–2.0% with a 4-KiB object. The added representation remains fixed or bounded instead of growing with the application object. Consequently, application content accounts for more than 92% of the encoded size at 1 KiB and more than

98% at 4 KiB. IPI therefore carries existing compact messages with a 5-B wrapper and adds the operation fields needed by CAV applications without materially increasing these larger objects.

The binding and experiment frames add path-specific fields beyond the representations in Table 5. The July 3–4 PC5 campaign reports complete serialized frame size. Its nonzero settings configure 256-, 512-, 1,024-, and 2,048-B IPI performance frames for submission to the vendor packet-data interface. Every retained nonzero sender record has a packet length equal to its configured frame size. The zero control retains the required IPI identity and timing fields and therefore produces a 51–55-B request. The PC5 analysis below varies this complete IPI frame size. The Uu analyses instead vary application-object size before IPI serialization, with encoded frame size retained in the sender records. These definitions separate the application object, IPI representation, and complete transmitted frame. The field campaigns then exercise IPI through the PC5 custom-data binding and the Uu TCP, MQTT, and UDP measurement adapters.

5.2 Direct PC5 Stationary Frame-Size Limits

The representation results establish the bytes added before a path binding. The PC5 experiments next measure how complete serialized frame size and field position affect the exchange across stationary points. Figure 6(a) compares response completion as the configured IPI frame size and stationary field point change. P1 is building-obstructed at 68 m; P2–P5 are 113, 212, 377, and 469 m from the RSU. The RTT panel uses a separate 7-m reference collection. In that collection, 0.25 KiB returned 999/1,000 responses; 0.5–2 KiB returned every response. Their p50 RTTs remained near 100 ms. Their p95 RTTs ranged from 106.694 to 111.663 ms (Figure 6(b)). These RTTs exceed the 10- and 25-ms cooperative budgets at the median and the 100-ms warning budget at p95.

The effect of frame size increased across the later field points. The 2-KiB failure rate rose from 0.2% to 99.2%, followed by 910 failures before the next collection was stopped.

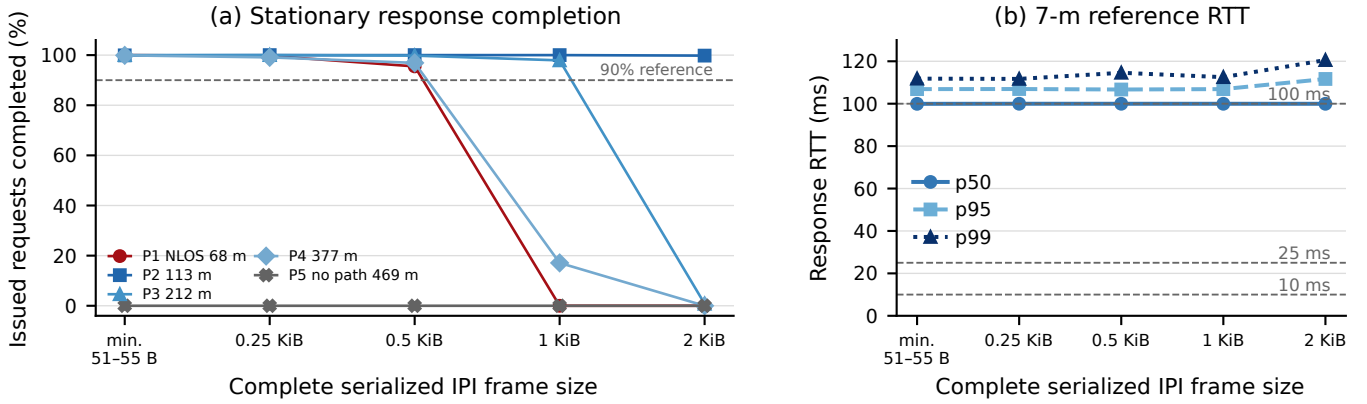


Figure 6: Stationary PC5 frame-size results. The response-completion rate is the percentage of issued requests that received a sequence-matched response before timeout. (a) Completion for the minimum-header control and complete serialized IPI frames from 0.25 to 2 KiB at P1–P5; P1 is the building-obstructed non-line-of-sight point. (b) Response p50, p95, and p99 RTT in a separate 7-m reference collection.

Over the same sequence, the 1-KiB failure rate rose from 0% to 2.1% and then 82.9%. The 0.25- and 0.5-KiB conditions remained available at more points. The final point returned no response for any configured frame size. The nearby non-line-of-sight (NLOS) point was obstructed by a building and is shown separately from the ordered field sequence.

5.3 5G Uplink: Payload Size and Deadlines

The PC5 measurements expose stationary frame-size and coverage limits for direct exchange. The 5G evaluation extends the payload range over the network-assisted Uu path. Each 5G application probe transmits an IPI cooperative-service workload frame from the vehicle. The d1 endpoint decodes each frame and immediately returns a compact correlated acknowledgment. Figure 7 compares workload responses at 100, 500, and 1,000 ms. Across the eight disjoint workload groups, the analysis contains 752,200 issued requests. Requests with payloads of at most 4 KiB returned all 60,000 responses. Only 37.76% arrived within 100 ms and 96.29% within 500 ms. The loaded 1-KiB group similarly achieved 24.12% at 100 ms and 93.47% at 500 ms. These conditions carry the payload sizes used by warnings, signal-priority coordination, and compact remote-recovery control. The fraction completed in time decreases as the deadline becomes shorter.

Larger payloads shift the envelope further. The 128–512-KiB group returned all 18,000 responses, with 0.99% within 100 ms and 88.36% within 500 ms. The group with payloads of at least 1 MiB also returned all 14,000 responses. None arrived within 500 ms, and 37.39% arrived within 1 s. These large vehicle-to-edge workload requests received their compact acknowledgments on timescales that exceed the interactive budgets in Table 3.

Field collections also changed the response tail. Compact-payload p95 RTTs remained within tens or hundreds of milliseconds, while 2-MiB p95 RTTs reached 2.5–10.8 s. A variable-path 256-KiB condition reached a 43.8-s p95. Payload size and field condition jointly shape the usable Uu envelope.

5.4 5G Uplink: Transport Effects on Completion and Latency

Payload size determines how much application data the uplink must carry; transport determines whether the encoded object is delivered as a stream or as one or more datagrams. We therefore compare TCP, MQTT over TCP, raw UDP, and UDP with adapter-level fragmentation at matched object sizes and field conditions. Figure 8 compares empty-payload controls, detector-derived payloads, and the 58.6-KiB stress payload. TCP and MQTT returned every response at each size in the –106-dBm collection. At –120 dBm, their 58.6-KiB p95 RTTs exceeded 660 ms, and their p99 RTTs exceeded 1 s.

The detector sweep also gives a conservative size boundary for a 500-ms CAV decision budget. Every evaluated TCP and MQTT condition through 19.2 KiB of application payload had a p95 RTT below 500 ms in both the –106- and –120-dBm collections. At 22.3 KiB, the –120-dBm MQTT p95 crossed the deadline at 539.780 ms. Later detector points fluctuated rather than restoring a monotonic size guarantee. These measurements place the 500-ms p95 ceiling for this uplink-heavy TCP/MQTT path at an approximately 20-KiB application payload before transport framing. The ceiling appeared with one UE on a dedicated 40-MHz n48 channel and no ambient traffic.

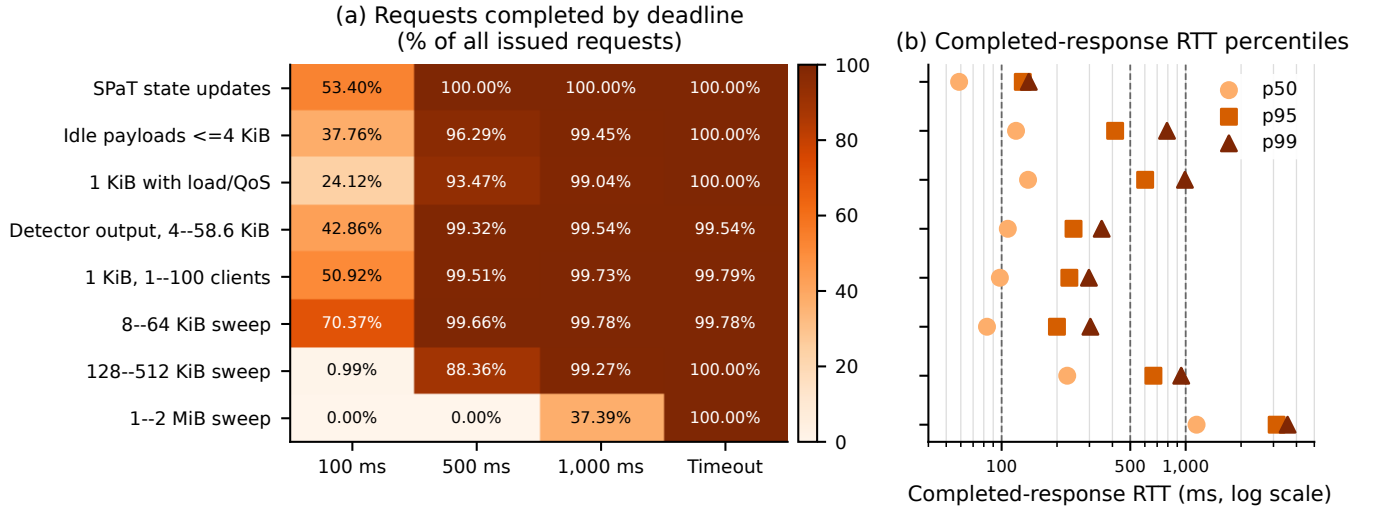


Figure 7: Uplink-heavy private 5G IPI payload-size and deadline results. The vehicle sends the stated workload payload, and d1 returns a compact acknowledgment after decoding it. (a) Percentage of all issued requests completed within 100, 500, and 1,000 ms and before timeout. Detector output denotes serialized object-detection result sizes from 4 to 58.6 KiB replayed as IPI payloads. (b) The p50, p95, and p99 RTT among completed responses.

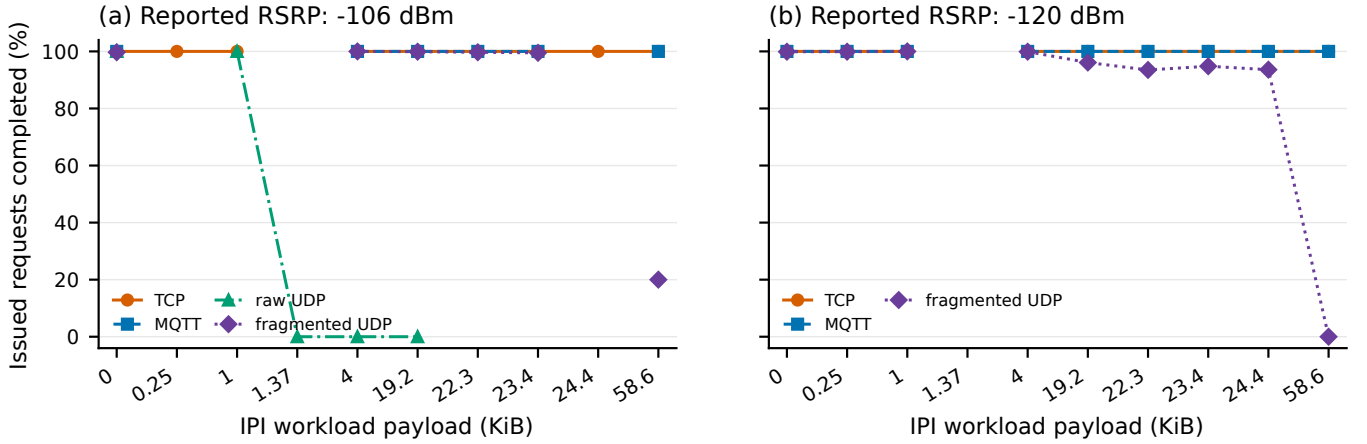


Figure 8: Uplink-heavy response completion by transport, IPI workload size, and reported RSRP. Each line reports the percentage of issued requests that returned a complete response. A gap denotes an unevaluated transport-size pair. Fragmented UDP uses 1,400-B adapter-level fragments and completes a response only after reassembly.

Raw UDP completed the empty and 1-KiB conditions but returned no response for the evaluated larger datagrams. Adapter-level fragmentation and receiver reassembly sustained at least 93.5% completion through 25 KiB. At 58.6 KiB, completion fell to 20% at -106 dBm and 0% at -120 dBm. TCP and MQTT use retransmission and buffering, while fragmented UDP loses more reassembled responses as payload size and path difficulty grow.

5.5 5G Uplink Across Stationary Radio Conditions

After separating transport effects, the stationary comparison holds the 70/20/10 profile and uplink-heavy exchange fixed while compact and detector-sized workloads run at four locations. Figure 9 plots the detector-sized results against the reported RSRP at each location. The -120, pooled -109 to -110, and -101 dBm blocks returned 5,994/6,000, 5,999/6,000, and 6,000/6,000 responses. The larger -109 to -110-dBm campaign returned 307,662/309,000 responses.

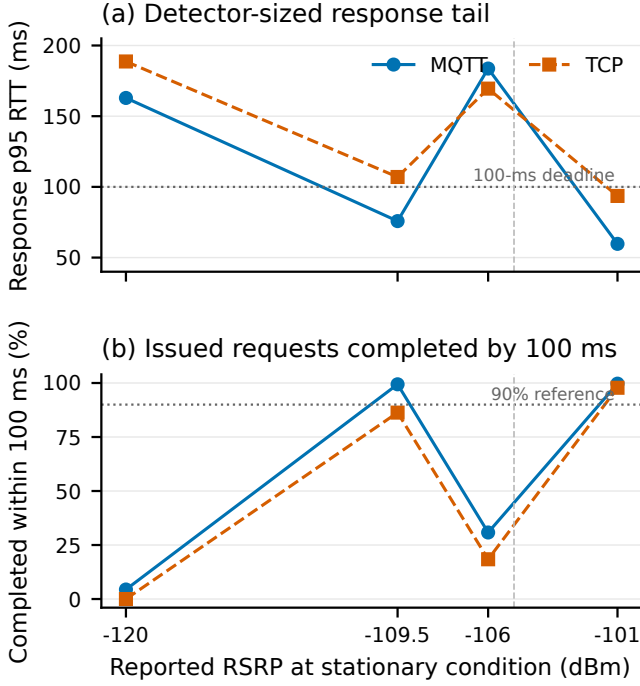


Figure 9: Detector-sized uplink-heavy responses across four stationary 70/20/10 conditions. (a) Response p95 RTT. (b) Percentage of all issued requests completed within 100 ms. The vertical line separates the weak and common/typical RSRP bins defined in Section 4.

Compact 1-KiB exchanges change little within each transport. Mqtt p95 spans 40.0–46.1 ms, and 99.1–100% of issued requests complete within 100 ms. TCP p95 spans 137.3–150.4 ms, with 0.5–4.6% completion within 100 ms. The detector-sized workload varies more. Mqtt 100-ms completion is 4.4%, 99.4%, 30.8%, and 99.7% at –120, –109 to –110, –106, and –101 dBm. TCP completion is 0%, 86.3%, 18.4%, and 97.8%. Across the same conditions, Mqtt p95 RTT is 162.9, 75.8, 183.7, and 59.7 ms; TCP p95 RTT is 188.7, 107.0, 169.5, and 93.6 ms.

The –101-dBm common/typical location produces the lowest detector-sized p95 and the highest 100-ms completion. The –106-dBm condition performs worse than the pooled –109 to –110-dBm conditions, so RSRP alone does not order the four outcomes. Payload size remains stable for the compact exchange and sensitive to the broader radio and path conditions for detector-sized traffic.

5.6 5G Downlink: Returned CAV Objects

The preceding campaigns place the larger application object on the uplink. We next reverse the large-object direction. A downlink-heavy CAV operation begins with a compact

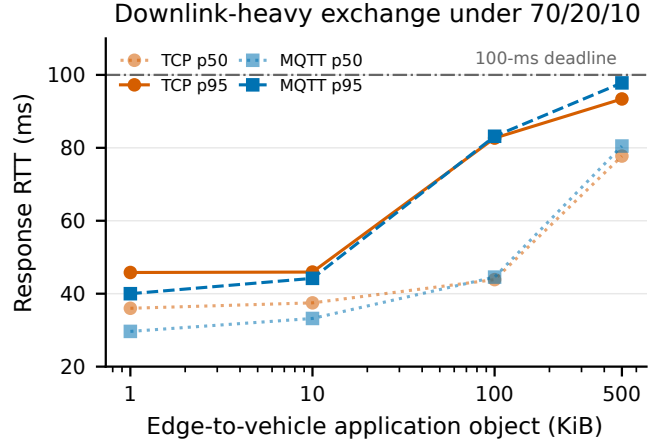


Figure 10: Downlink-heavy CAV object latency under the established 70/20/10 profile. The vehicle sends a compact request and validates the complete 1–500-KiB edge-to-vehicle object. Each point summarizes 500 exchanges.

vehicle request and ends after the vehicle receives and validates the returned application object. This pattern represents raw sensor data, processed perception, planning data, control data, or other state requested from an edge application. Under the established 70/20/10 profile, the GL-X3000 downlink campaign returned all 4,000 TCP and all 4,000 Mqtt responses across 1, 10, 100, and 500 KiB.

Figure 10 shows p95 RTT rising from 45.8 to 93.4 ms for TCP and from 40.0 to 97.8 ms for Mqtt as the returned object grows from 1 to 500 KiB. The 500-KiB response tails remain below the 100-ms reference in this stationary 70/20/10 condition. This established-profile result provides the baseline for testing whether a different radio configuration preserves burst latency and sustained-transfer goodput in both directions.

5.7 5G TDD and Gateway Effects

The TDD study evaluates whether the 40/40/20 behavior persists across vehicle gateways. The MG52 campaigns measured both profiles, but route recovery, serving-cell changes, and weaker signal accompanied part of the 40/40/20 collection. We therefore repeated the complete bidirectional matrix with a GL-X3000 at one stationary location. Within this repeat, TDD is the changing configuration; gateway, location, application, transport, and payload matrix remain fixed.

The GL-X3000 repeat contains 8,000 validated exchanges under each profile. Every exchange returned a response. The profile difference appears in latency rather than eventual completion.

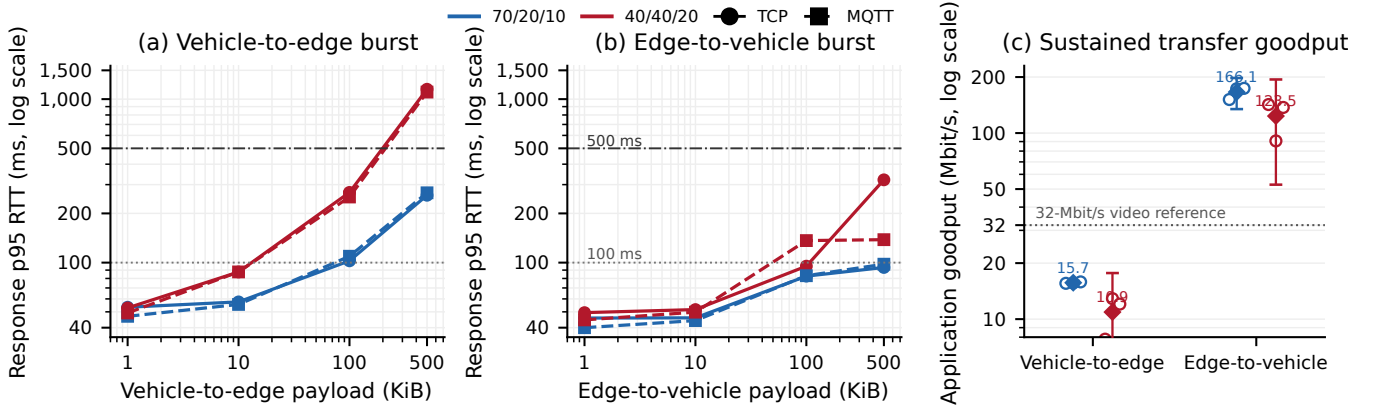


Figure 11: Same-device performance for CAV bursts and sustained transfers. The GL-X3000 blocks use the same stationary location; 70/20/10 and 40/40/20 have median RSRP of -99 and -98 dBm and median SINR of 20 dB. Panels (a–b) combine TCP and MQTT p95 RTT; each profile-transport marker summarizes 500 validated exchanges. The dotted and dash-dotted lines mark 100- and 500-ms application references. Panel (c) combines three exact 50-MiB transfers per profile and direction on a logarithmic goodput axis; open circles are individual transfers, diamonds are means, and whiskers are t -based 95% confidence intervals. The dotted line marks the 32-Mbit/s remote-teoperation video reference.

At 1 KiB, vehicle-to-edge p50 RTT was effectively tied. The difference grew with the vehicle-originated object. At 500 KiB, 40/40/20 increased TCP p50 RTT from 231.046 to 609.652 ms. TCP p95 increased from 258.200 to 1,149.368 ms. MQTT p50 increased from 238.607 to 580.942 ms, while p95 increased from 267.336 to 1,104.331 ms. The corresponding p95 ratios were 4.45 and 4.13. Edge-to-vehicle objects also had higher p95 RTT under 40/40/20 at every evaluated size and transport. The 500-KiB ratios were 3.44 for TCP and 1.41 for MQTT (Figure 11(a–b)).

These tail changes reduced deadline completion across the equally sized workload matrix. Under 70/20/10, 14.54%, 0.04%, and 0% of the 8,000 exchanges exceeded 100, 500, and 1,000 ms. Under 40/40/20, the corresponding fractions were 30.85%, 9.81%, and 1.26%. The latter profile produced 101 second-scale exchanges; 82 occurred in the two 500-KiB vehicle-to-edge conditions. Thus, the alternative configuration widened the large-burst tail even though it nominally assigns twice as much of the frame to fixed uplink use.

The radio measurements place both profile blocks in the same signal category. Median RSRP was -99 dBm under 70/20/10 and -98 dBm under 40/40/20. Median SINR was 20 dB in both blocks. Restricting the analysis to their common RSRP/SINR range retained 15,153 exchanges and the same latency ordering in 14 of 16 p50 comparisons and 15 of 16 p95 comparisons. At 500 KiB, TCP vehicle-to-edge server processing remained 1.617–1.619 ms at p50 while response RTT changed by 378.606 ms. The latency ordering therefore persists within common measured signal support, while the endpoint processing time remains nearly identical.

The exact transfers measure the sustained-stream component of CAV traffic. Each profile completed three 50-MiB uploads and three 50-MiB downloads with matching byte counts and checksums. Vehicle-to-edge goodput averaged 15.691 Mbit/s under 70/20/10. It averaged 10.913 Mbit/s under 40/40/20, a 30.4% reduction. Edge-to-vehicle goodput averaged 166.106 and 123.451 Mbit/s, a 25.7% reduction. Every 40/40/20 transfer was slower than every corresponding 70/20/10 transfer (Figure 11(c)). The two upload means provide 49.0% and 34.1% of the 32-Mbit/s remote-teoperation video reference in Table 3.

The cross-gateway controls separate device improvement from the profile ordering. Under 70/20/10, GL-X3000 upload and download goodput exceeded the closest MG52 control by 30.8% and 31.0%. The earlier MG52 large-uplink collapse did not recur with the GL-X3000, but the same-device comparison still favored 70/20/10 for burst latency and sustained goodput. Earlier profile transitions also changed route, cell state, and serving cell. Later matched blocks completed their exchanges but retained configuration-dependent latency and goodput. In this deployment, the evaluated 40/40/20 implementation was not ready for deadline-sensitive CAV traffic. Its initial MG52 rollout was operationally unstable, and its matched GL-X3000 repeat produced higher burst latency and lower directional goodput than 70/20/10. The additional nominal uplink allocation did not translate into predictable application performance.

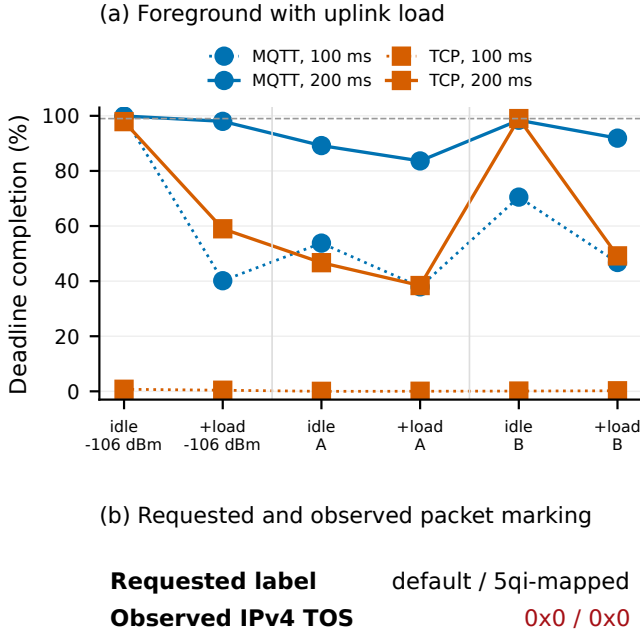


Figure 12: Compact foreground exchanges during a sustained uplink stream. (a) Percentage of issued 1-KiB requests completed within 100 and 200 ms at idle and with the uplink generator active; Collections A and B are separate weak-signal field collections. (b) Requested application labels and the observed vehicle-side Internet Protocol version 4 Type of Service (TOS) value.

5.8 Concurrent Bursts and Sustained Streams

The configuration comparison uses one application flow at a time. During remote teleoperation, a vehicle can sustain a video stream while faults, control updates, or planning requests create additional deadline-sensitive exchanges. The load experiment models this communication pattern with a continuing vehicle-to-edge stream and a compact 1-KiB foreground exchange on the same host and UE. With the generator active, MQTT C(100 ms) fell from 99.70% to 40.15%, while TCP C(200 ms) fell from 97.90% to 59.00%. Across the load collections, achieved background traffic ranged from 1.320 to 5.743 Mbit/s. The active generator reduced foreground deadline completion before reaching the offered rate.

The response tails show the same shift. In the -106-dBm repetitions, the active generator increased MQTT p95 from 52–60 to 174–176 ms and TCP p95 from 162–194 to 388–420 ms. The response tail did not vary monotonically with the number of streams, so achieved background rate alone does not predict foreground latency.

The default and 5qi-mapped labels both produced observed TOS values of 0x0 in the vehicle capture (Figure 12(b)). The foreground and background flows entered the measured path without an observed packet-class difference between the requested labels. The sustained stream therefore displaced a compact exchange from its deadline before reaching the 32-Mbit/s video reference, and the requested application label did not create observable packet differentiation on the vehicle interface.

5.9 Concurrent Application Demand

The preceding experiment combines one continuing stream with one foreground exchange. CAVs can also start several perception, planning, and control operations at the same time. The client-demand experiment increases the number of simultaneous application flows behind the same physical UE.

The client-demand experiment scales a homogeneous 1-KiB exchange across multiple application clients on one host and UE. At 100 clients and -106 dBm, aggregate p95 RTT was 81.850 ms for MQTT, 264.070 ms for TCP, and 113.391 ms for UDP. At -120 dBm, MQTT and TCP p95 rose to 5,135.984 and 2,908.420 ms. UDP p95 was 172.500 ms, but 9,189 of 100,000 cycles received no response. Figure 13(c–d) shows that deadline completion falls as client demand grows. Per-client tail distributions further show that individual flows can miss the 100- and 200-ms budgets even when most aggregate requests eventually receive a response.

5.10 Endpoint Interruption and Recovery

The demand experiment varies concurrent flows while each endpoint remains available. To measure recovery separately, the interruption experiment restarted a receiver or broker for 10 s while 1-KiB requests continued. MQTT broker, MQTT receiver, TCP receiver, and UDP receiver restarts returned 955/1,000, 991/1,000, 953/1,000, and 969/1,000 responses. At the 200-ms remote-assistance budget, completion was 94.7%, 97.7%, 92.9%, and 96.9%. The gaps between received responses were 15.1007, 13.3674, 11.7147, and 13.4635 s. Every response gap exceeds the injected outage and is absent from response-only latency percentiles. Among the responses that arrived, the corresponding p99 RTTs were 90.553, 259.906, 361.228, and 63.027 ms.

5.11 Cross-Application Comparison

The preceding subsections report path-specific response latency and deadline completion. Figure 14 places the p50, p95, and maximum RTT of each selected condition beside its application deadline. Each response marker is green when it meets the deadline and red when it exceeds the deadline.

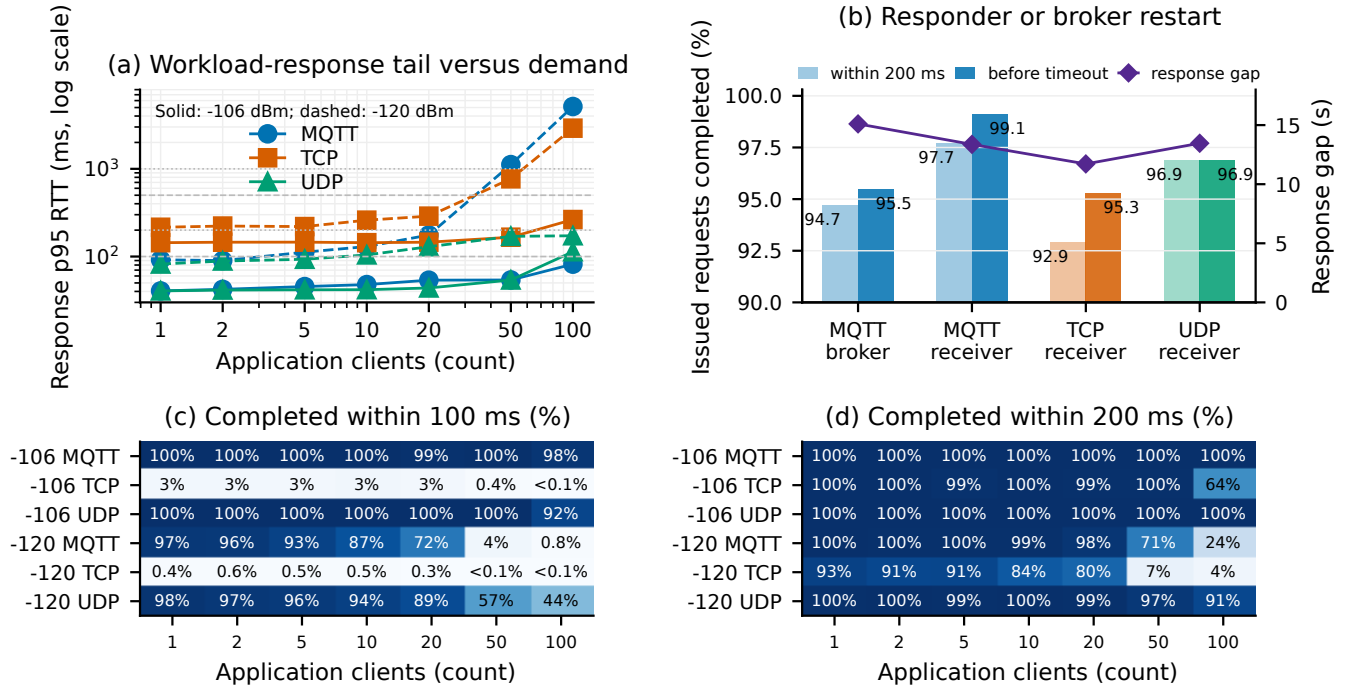


Figure 13: Client-demand and interruption results for 1-KiB application payloads. (a) Response p95 RTT in milliseconds versus client count; solid and dashed lines mark the -106- and -120-dBm weak-band collections. (b) Completion percentage within 200 ms and before timeout, together with the response gap in seconds after a 10-s restart. (c-d) Percentages of all issued requests completed within 100 and 200 ms.

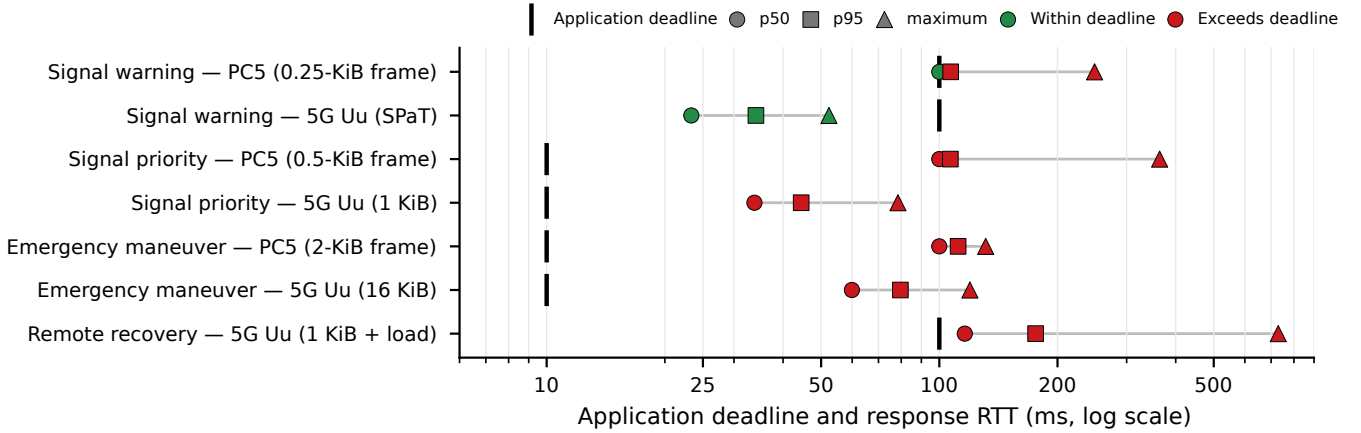


Figure 14: Application deadlines and measured response distributions. Each black tick is the deadline from Table 3. Circles, squares, and triangles show response p50, p95, and maximum RTT; green markers meet the deadline and red markers exceed it. Row labels state the path and workload size used for each comparison. PC5 row labels report complete serialized IPI frame size; Uu row labels report application-object size.

The 5G MQTT SPaT exchange has p50, p95, and maximum RTTs of 23.4, 34.1, and 52.4 ms, so its complete measured distribution remains within the 100-ms signal-warning deadline. The stationary 0.25-KiB PC5 frame exchange has a 100.0-ms

p50, a 106.9-ms p95, and a 248.6-ms maximum; only its median remains within the same deadline. Both signal-priority conditions and both emergency-maneuver conditions exceed

their 10-ms deadlines at p50. The loaded 1-KiB remote-recovery condition also exceeds its 100-ms deadline at p50, p95, and maximum, which reach 116.1, 175.9, and 729.8 ms.

Operating conditions also alter the same workload when payload and deadline are held fixed. For the detector-sized workload under 70/20/10, 100-ms MQTT completion is 4.4% at -120 dBm. It reaches 99.7% at the common/typical location with -101 dBm recorded RSRP. The -106 -dBm 70/20/10 condition reaches 30.8%. Thus, the same profile and workload cross the application deadline as location changes.

The measured deadline outcome also differed between the two deployed configurations within the same signal category. In the GL-X3000 comparison, the fraction of all exchanges exceeding 500 ms increased from 0.04% under 70/20/10 to 9.81% under 40/40/20. The latter configuration also reduced sustained upload and download goodput by 30.4% and 25.7%. The corresponding RSRP medians differed by 1 dB, and both blocks had 20-dB median SINR.

Shared demand changes the same deadline a second way. Compact 1-KiB MQTT completion falls from 99.7% to 40.15% with competing uplink traffic (Figure 12(a)) and from 96.6% to 0.751% as -120 -dBm client demand grew from one to 100 (Figure 13(c)). The default and 5qi-mapped labels both produced vehicle-side TOS 0x0. Consequently, the measured path exposed no packet-class distinction between these labels as competing traffic and concurrent application demand moved the same 1-KiB workload across its deadline.

5.12 Summary and Insights

The cross-application comparison identifies different payload, coverage, direction, radio-configuration, and demand limits. CAV traffic can begin with an event-triggered uplink or downlink burst and then continue as a sustained stream during remote teleoperation or continuing sensor delivery. Other deadline-sensitive operations can begin while that stream remains active. Increasing the workload and varying the operating conditions reveal where each path crosses the application deadline and how many issued requests remain within it. These measured boundaries lead to three insights:

- 1) **The measured commercial direct V2X path supports compact J2735 messages only within a limited payload and coverage envelope.** The 68-m building-obstructed point lost every 1- and 2-KiB exchange, whereas the 113-m point completed 99.8–100% of each tested frame size. Deployment planning must therefore account for obstruction and road geometry when placing RSUs or supplemental paths.
- 2) **The measured 5G uplink supports only severely limited CAV packet sizes within decision deadlines.** The conservative 500-ms p95 ceiling was approximately 20 KiB, and only 37.76% of issued requests

at or below 4 KiB completed within 100 ms. These limits appeared on a dedicated 40-MHz channel with one UE and no ambient contention. Because matched large objects had much longer uplink than downlink tails, CAV developers need direction-specific packet budgets and must reduce or progressively transmit vehicle-originated state when necessary.

- 3) **5G/6G systems must support event-triggered CAV bursts, sustained directional streams, and concurrent deadline-sensitive traffic.** In the same-device comparison, 40/40/20 produced more deadline misses and lower sustained goodput than 70/20/10 despite its larger nominal uplink allocation. A sustained uplink stream also moved compact foreground exchanges across their deadlines. Radio vendors and mobile network operators must validate balanced and uplink-oriented configurations across radios and vehicle gateways, while schedulers isolate simultaneous flows by direction, deadline, and duration.

6 Future Research Directions

Future CV and ITS deployments should choose RSU spacing, antenna geometry, and supplemental coverage from the fraction of compact J2735 messages completed within the application deadline. The evaluation must include urban intersections, high-rise street canyons, highways, and sparsely served roads because distance alone does not characterize obstruction or coverage.

At the matched large-object condition, vehicle-originated content required substantially longer than the same-sized edge-returned content. CAV developers should therefore select, compress, partition, or progressively transmit vehicle-originated state according to the available uplink, the current radio condition, and the decision deadline. These mechanisms also need explicit expiration and recovery so that delayed fragments do not consume resources after the application can no longer use the complete object.

For mobile network operators and 5G/6G researchers, CAV demand changes in arrival time, dominant direction, and duration. A driving event can create an immediate uplink- or downlink-heavy burst. Remote teleoperation can then sustain video, state, and control traffic for the remainder of the assistance session. Other deadline-sensitive operations may begin while this stream remains active. The same-device measurements show that a nominal TDD ratio does not determine end-to-end burst latency or sustained goodput. The load experiments also show that a continuing stream can move compact foreground exchanges across their deadlines. Future radio work should make balanced and uplink-oriented frame structures interoperable across chipsets, vehicle gateways, and infrastructure while preserving attachment and

service state. Schedulers must consider each workload’s direction, deadline, and expected duration while isolating simultaneous traffic. Common benchmarks should combine asynchronous bursts with sustained streams across independently scheduled UEs while varying signal condition, TDD allocation, mobility, and handover. They should report direction-separated RTT, deadline completion over all issued requests, upload and download application goodput, fairness, configuration-state provenance, and recovery gaps [22, 27].

7 Conclusion

This paper presents a real-vehicle study of whether today’s direct V2X and network-assisted 5G paths can support compact CV/ITS messages and correlated CAV operations. The study measures response latency, deadline completion, and directional goodput as payload, direction, radio condition, configuration, and shared demand change. To make the two workload classes comparable, we also contribute IPI, an application-layer protocol that preserves existing J2735 messages and represents correlated CAV operations through one application contract, distinct message and operation profiles, and path-specific bindings.

The measurements establish different operating limits for the two paths. The commercial PC5 path carried compact J2735 traffic at favorable stationary points, but larger serialized frames failed first as coverage deteriorated. The 5G path carried larger stateful exchanges, but payload size and network conditions determined whether vehicle-originated requests completed within the application deadline. Same-device tests also showed that a larger nominal uplink allocation did not guarantee lower burst latency or higher directional goodput, while sustained and concurrent traffic displaced compact exchanges from their deadlines. Application-ready CAV communication therefore requires compact representations, stable bidirectional radio operation, and scheduling that responds to each exchange’s direction, duration, and deadline.

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A IPI Reference Schema

Table 6 separates the application objects from their serialized profiles. Message-mode calls preserve typed or pre-encoded J2735 content. Operation-mode calls add the association and outcome fields required by a continuing CAV interaction. The runtime stores active sessions and responses, while each binding serializes the fields required by its path.

The C++17 implementation maps supported J2735 structures into the IPI application contract and preserves externally encoded J2735 messages byte for byte. Multi-byte integers in the deterministic operation and session encodings use network byte order. Length-prefixed fields are checked before allocation, and decoders reject truncated records, unsupported enumerations, invalid field bounds, and trailing bytes. The cooperative-operation expiration field uses tenths of a second since the epoch. Session leases use a monotonic clock, so a wall-clock adjustment cannot extend a lease. The measurement adapters record wall-clock fields for trace alignment and use the sender’s monotonic clock for RTT.

The tests cover typed and pre-encoded J2735 content, independently generated conformance vectors, operation roles, session lifecycle, correlation, malformed input, PC5 framing, and broker-backed exchange. The runtime validates session state, request identity, correlation, expiration, and terminal-response behavior. Applications and deployments define domain-specific transition policy, security, admission, radio scheduling, and path selection.

B Deployment and Measurement Details

Table 7 expands the deployment context from Section 4. The AirSpeed inventory records versions 22.0-24.0.0 and 22.00-53-0.0. Its two enabled n48 cells map to two sector carriers. The primary application traffic used 70/20/10 and Cell 2. The August 15–19 TDD study records application direction, serving cell, location, reported signal context, and both cells’ administrative state. Seven overlapping DU Cell exports provide five-minute cell counters for the available experiment windows. Their zone-less timestamps are interpreted as EDT from traffic alignment. The CBRS export records Category B General Authorized Access operation, one requested and authorized grant, no suspended grant, and a successful Spectrum Access System heartbeat.

One CAV gateway serves as the sole physical experiment UE. The dedicated radio and 40-MHz channel operate in a clean n48 band without ambient contention. The load experiment places one to four streams behind it. The client-demand experiment places 1–100 closed-loop clients behind the same attachment. These workloads traverse the gateway, Uu path, private core, transport, and d1 responder. The matrix excludes public-network handover and roaming. The controlled band isolates experimentally introduced load and

application demand from ambient radio traffic. Profile transitions can still change route, cell state, serving cell, signal, and transport state, so those fields remain part of each comparison.

The 5G radio survey comprises three continuous drives through the testbed. After removing 123 records from a frozen radio-value interval, it retains 1,178 band-48 records with a median RSRP of -108 dBm and a median recovered NR SINR of 9 dB. We use this survey for route-level coverage and retain the signal records collected with each stationary application campaign for its workload comparisons. The PC5 collections include a reference point, a building-obstructed NLOS point, successive stationary field points, and four driven routes. Exact GNSS coordinates are transformed to a local frame in the analysis artifacts.

C Workload and Experiment Matrix

The July 3–4 PC5 stationary sweep configures complete serialized IPI performance frames at 0.25, 0.5, 1, and 2 KiB, normally for 1,000 requests per size at 100-ms intervals and a 1,000-ms timeout. The nonzero settings pad both the request and response to the configured frame length. A separate zero control produces a 51–55-B request because the identity and timing header remains present. Each response copies the request’s origin, sequence, initial send time, and configured length. Timeout-dominant collections retain every transmitted request. The four route collections use 0.25-KiB serialized IPI frames at 200-ms intervals. One stopped route contains 675 attempts; the other routes contain 1,000 attempts each.

The five stationary uplink-heavy 5G payload campaigns were collected on May 13, 14, 15, 21, and 22, 2026. Most completed conditions contain 1,000 requests at 200-ms intervals. Every application request uses an IPI cooperative-service frame. The requested size denotes the application payload before IPI serialization; sender logs retain the encoded IPI frame size. The dataset manifest contains 22,431 artifacts from OpenDAIR-V2X, V2X-Seq, TruckV2X, and V2X-Radar. Images span 25.5–666.9 KiB, point clouds span 15 KiB–3.85 MiB, and maps span 22.3–44.9 MiB. These ranges motivate the fixed-size sweeps. The 128–512-KiB and 1–2-MiB transfers represent chunks drawn from these application-payload scales.

We derive payload sizes from 922 saved V2X-Radar validation outputs. The model assigns a 256-B header and 96 B per predicted object. Payloads range from 12.9 to 24.4 KiB, with median, p95, and p99 sizes of 19.2, 22.3, and 23.4 KiB. Network replay fills an IPI cooperative-service content field to each representative size. The 58.6-KiB condition is a separate stress payload. The responder returns a compact correlated acknowledgment immediately after decoding the frame.

Table 6: IPI application objects and serialized profiles.

Component	Principal fields or encoding	Function
Application API metadata	Message, source, context, send time, transport class, and optional session, correlation, sequence, priority, and expiration fields	Supplies application identity and context; a serialized profile selects the fields required by its exchange
Compact J2735 content profile	One-byte message type, four-byte big-endian payload length, and complete J2735 bytes	Carries typed BSM, PSM, MAP, SPaT, SRM, and SSM content or preserves a pre-encoded J2735 message in $P + 5$ B
Correlated CAV operation profile	Session and vehicle identifiers, service class, request/update/completion/rejection role, option flags, expiration, confidence, and length-delimited typed or opaque content	Associates state and outcomes with a perception, planning, control, or map operation
Runtime session record	Vehicle profile, role, requested operations, state, monotonic lease, heartbeat interval, telemetry, and responses	Maintains local state for registration, heartbeat, invocation, response retrieval, and termination

Table 7: Detailed radio, gateway, and endpoint configuration.

Element	Retained configuration	Measurement record
5G radio	Dedicated AirSpeed 2900; NR n48; TDD; 40 MHz; 30-kHz control and synchronization-signal subcarrier spacing; 20-ms synchronization block period	Clean n48 band without ambient traffic; one uplink and up to two downlink layers configured; downlink 256-quadrature-amplitude modulation enabled as a cell capability
Cells	Cell 1: absolute radio-frequency channel number 637992 and synchronization signal at 3558.720 MHz; Cell 2: 645334 and 3669.600 MHz	Primary traffic and the matched August 17 blocks used Cell 2; the second August 16 40/40/20 TCP repetition used Cell 1; the 70/20/10 Cell 2 lock was operator-reported
Power and antenna	33-dBm cell transmit power; 34 dBm/MHz effective isotropic radiated power; 17-dBi cross-polarized integrated antenna	Product pattern approximately 65° azimuth by 8° elevation; structural centerline 20 ft above roof level
Duplex configuration	Established application profile: 70/20/10 downlink/uplink/dynamic frames; August 15–19 comparison: 40/40/20 and 70/20/10; fixed 10D4G packing	Route, cell state, and serving cell changed during the August 15–16 transition; August 17 and 19 supplied matched workload comparisons; separate 60/20/20 runs are excluded; the attempted 30/60/10 configuration produced no valid experiment
Vehicle gateway	Cisco Meraki MG52-HW campaign gateway with Telit FN990A40 modem and internal antenna; GLiNet GL-X3000 replacement gateway; 1-Gbit/s full-duplex Ethernet	The August 19 GL-X3000 repeat records gateway RSRP, Reference Signal Received Quality (RSRQ), and SINR while evaluating both TDD profiles at one stationary location
Session and endpoint	SIM provisioned on Data Network Name cisco5g; operator-reported default 5G QoS Identifier 9; on-site MX250 and directly connected d1 host	Host captures observed TOS 0x0 for default and 5qi-mapped experiment labels
Field context	Sole experiment UE; separate three-drive survey with a Samsung 22 handset running G-NetTrack Pro; 1,178 valid band-48 records; RSRP –121 to –88 dBm; recovered NR SINR –20 to 30 dB	Drive-survey medians: –108 dBm RSRP and 9 dB SINR; August 17: operator-reported –100-dBm RSRP and –13-dB RSRQ; August 18: –105 to –115 dBm; August 19 gateway medians: –99 and –98 dBm RSRP and 20-dB SINR
PC5 hardware	Mocar OBU, HUALI RSU, 2020 vendor J2735 stack, packet type 0x1b, 4,080-B installed-interface limit	GNSS-joined application outcomes; no received-power, channel, resource-pool, modulation, or retransmission export

The August 15–16 TDD runs retain the same uplink-heavy timing contract. The August 17 matched experiment instead uses a compact vehicle request followed by an exact 1-, 10-, 100-, or 500-KiB edge-to-vehicle response. Each profile contains 1,000 validated TCP and MQTT exchanges per payload. The vehicle timer includes complete payload receipt and sequence, length, and checksum validation. The same experiment then performs ten exact 50-MiB TCP uploads and ten exact downloads per profile. Endpoint byte counts, checksums, and receiver-observed durations define application goodput.

The August 18 40/40/20 repeat uses 1,000 uplink-heavy and downlink-heavy exchanges at 1, 10, and 100 KiB. Each transport contains 250 declared uplink-heavy exchanges at 500 KiB and 500 downlink-heavy exchanges. Reported

RSRP changes from –105 to –110 and then –115 dBm during the run. Its shortened bulk stage contains one complete upload/download pair and one additional upload. The failed August 17 downlink preflight, the bulk pilot, and the unintended August 18 partial preflight remain diagnostic artifacts outside the primary estimates. The August 19 GL-X3000 repeat uses 500 validated exchanges for every combination of direction, transport, profile, and 1-, 10-, 100-, or 500-KiB payload. It adds three exact 50-MiB uploads and three exact downloads per profile. The two profile blocks retain separate gateway radio records and use the same stationary location. Table 8 records the controlled conditions and the corresponding measurement for every experiment family.

In the closed-loop application-demand collection, each application client waits for a response or timeout before the

Table 8: Complete workload and communication experiment matrix.

Question	Controlled conditions	Recorded outcome
Message and compact exchange	J2735 callback checks; local IPI tests; decoded PC5 IPI performance frames with sequence-correlated responses; Uu 0–4-KiB IPI frames over TCP/MQTT	Decode or callback result, response completion, RTT
Payload scale	PC5 minimum-header control and 0.25–2-KiB complete IPI frames; uplink-heavy Uu IPI frames with 0–2 MiB vehicle-originated content; downlink-heavy Uu frames with 1–500 KiB returned content; detector-derived sizes; 58.6-KiB stress payload	$C(D)$ and response p50/p95/p99
Field and route coverage	PC5 stationary points and four routes; original Uu stationary collections and Samsung 22 continuous-drive survey; matched August 17 location 3; August 18 time-segmented context; August 19 same-location gateway measurements	GNSS-joined response completion, RTT, and separate radio context
Protocol formation	TCP, MQTT QoS 0 over TCP, raw UDP, and UDP with 1,400-B adapter-level fragments	Returned acknowledgment after decode or returned payload after reassembly and validation
TDD and application direction	August 15–16 uplink-heavy repetitions; August 17 matched downlink-heavy responses and exact transfers; August 18 weak-signal direction repeat; August 19 same-device 1–500-KiB bidirectional and exact-transfer matrix under both profiles	Profile-transition repeatability, response RTT, deadline completion, exact application goodput, device sensitivity, and radio context
Exact directional transfer	Ten exact 50-MiB uploads and downloads per August 17 profile; three per August 19 profile and direction; receiver byte-count and checksum validation	Individual transfer goodput, mean, and 95% confidence interval
Competing traffic	Separate historical 15-minute offered controls; 1-KiB foreground with one to four offered 25-Mbit/s uplink streams	Achieved background rate and foreground deadline completion
Application demand	1, 2, 5, 10, 20, 50, and 100 clients; 1-KiB closed-loop requests; sole experiment UE	Aggregate and per-client response completion and RTT
Packet marking	Default and 5qi-mapped application labels	Vehicle-interface Internet Protocol header capture
Interruption	TCP receiver, UDP receiver, MQTT receiver, or MQTT broker restarted for 10 s	Missing attempts, deadline completion, and response gap

next request from that client and then applies a 200-ms interval. At negligible response time, five requests/s/client is the maximum nominal rate. A locked-cell 70/20/10 collection additionally uses four conditions: one idle 1-KiB client, one idle 23.4-KiB client, the 1-KiB client with offered uplink traffic, and 100 1-KiB clients. TCP, MQTT, and UDP occupy separate five-minute collection bins with two repetitions. The first three conditions use 500 probes per repetition; the 100-client condition uses 500 probes per client per repetition.

Every issued primary request/response attempt remains in its condition denominator, including missing, timed-out, late, and failed attempts. Entirely excluded diagnostic roots and never-started bulk repetitions do not enter a primary estimate. Operator-shortened conditions use their declared retained counts. The request/response experiments carry application-derived workloads in IPI cooperative-service frames and measure RTT and deadline completion. PC5 uses decoded IPI performance requests and responses matched by origin and sequence identifiers. Exact bulk transfers instead use validated bytes, checksum, receiver duration, and application goodput.

D Detailed Results for Additional Conditions

Several experiment families include conditions that are not represented by dedicated figure panels. This section reports their complete numerical outcomes.

D.1 Direct PC5 Route Results

Each route used a complete 0.25-KiB serialized IPI request. Routes 1 and 2 used a 1,000-ms timeout, while Routes 3 and 4 used a 500-ms timeout. The response percentiles include only sequence-matched responses; completion uses all issued requests as its denominator.

Across the four routes, 59.1–76.3% of issued requests received a response. The p95 RTT among received responses was below 50 ms on three routes even though 26.9–40.9% of all requests on those routes timed out. The GNSS join places missing responses in recurring no-response segments.

D.2 Field-Specific Uplink Payload and Transport Results

Field collections changed the response tail across both payload and transport. In one collection, 0–4-KiB p95 RTT was 36–49 ms for MQTT and 100–140 ms for TCP. At 2 MiB, p95 reached 2,536 and 3,633 ms. At another location, 2-MiB TCP p95 reached 10,767 ms. In a variable-path collection, empty-payload TCP and MQTT p95 rose to 1,142 and 510 ms. The 256-KiB MQTT condition returned 999/1,000 responses with a 43,836-ms p95.

The transport comparison used empty-payload controls, detector-derived payloads, and the 58.6-KiB stress payload. In the –106-dBm collection, TCP and MQTT returned every response at each size. MQTT p95 RTT increased from 49.826 ms with an empty payload to 183.705 ms for the 23–24-KiB group and 403.790 ms at 58.6 KiB. The corresponding

Table 9: Direct PC5 route response completion and RTT.

Route	Attempts	Responses	Completion (%)	Timeout (ms)	p50 RTT (ms)	p95 RTT (ms)	p99 RTT (ms)
R1	675	515	76.3	1,000	99.610	108.421	112.436
R2	1,000	731	73.1	1,000	28.326	41.887	55.880
R3	1,000	591	59.1	500	29.204	43.923	54.317
R4	1,000	716	71.6	500	28.276	41.621	50.651

TCP values were 151.696, 169.540–213.894, and 381.409 ms. In the –120-dBm collection, 58.6-KiB MQTT and TCP p95/p99 increased to 770/1,230 and 662/1,027 ms.

Raw UDP returned 1,000/1,000 empty-payload and 300/300 1-KiB responses. It returned 0/45, 0/100, and 0/100 at 1.37, 4, and 19.2 KiB. Application fragmentation and receiver reassembly increased response completion at –106 dBm. Completed responses reached 99.5–100.0% through 23.4 KiB, while the 58.6-KiB condition returned 20/100 responses. At –120 dBm, payloads through 4 KiB returned 99.9–100.0%, 19–25-KiB payloads returned 93.5–96.1%, and 58.6 KiB returned 0/100.

D.3 Gateway and TDD Replication Results

At 70/20/10, the closest MG52 control and the GL-X3000 repeat had reported or measured median RSRP of –100 and –99 dBm. The GL-X3000 repeat measured 15.691 Mbit/s upload and 166.106 Mbit/s download goodput. These rates are 30.8% and 31.0% above the MG52 control rates of 12.000 and 126.786 Mbit/s. Under 40/40/20, the earlier MG52 collection crossed weaker reported signal levels from –105 to –115 dBm. The later GL-X3000 collection at median –98 dBm measured 10–500-KiB vehicle-to-edge p50 RTTs 63.7–92.4% below the earlier MG52 collection, so the earlier large-uplink collapse did not recur. The same-device comparison still produced lower burst latency and higher sustained goodput under 70/20/10. The higher rates in the replacement-device run therefore did not remove the measured gap between the two deployed configurations.

Consecutive 70/20/10 runs changed TCP p95 RTT by only 0.22–1.29% across 1–100 KiB. During the initial 40/40/20 rollout, the route required recovery, cell administrative state changed, and separate repetitions selected different serving cells. A later matched MG52 downlink block completed all 8,000 exchanges, although its payload and transport effects differed. The GL-X3000 repeat completed both directional matrices but retained the large-object latency and goodput gap.

D.4 Sustained-Stream Load and Packet-Marking Results

The two –106-dBm repetitions achieved 1.320–2.267 Mbit/s of background traffic. MQTT C(100 ms) fell from 99.70% at idle to 40.15% with the generator active. At 200 ms, it changed

from 99.95% to 98.00%. TCP achieved 0.75% and 0.40% at 100 ms. Its 200-ms completion fell from 97.90% to 59.00%. Across one, two, and four background streams, achieved throughput ranged from 1.990 to 5.743 Mbit/s. The pooled active-generator conditions in two weak-band collections produced MQTT/TCP 200-ms completion of 83.63%/38.40% and 91.90%/49.17%.

At –106 dBm, the active generator increased MQTT p95 from 52–60 to 174–176 ms. TCP p95 increased from 162–194 to 388–420 ms. Across the one-, two-, and four-stream sweep, MQTT p95 was 170, 468, and 182 ms. TCP p95 was 391, 402, and 410 ms. The MQTT tail did not vary monotonically with the number of streams.

The default and 5qi-mapped labels both produced observed TOS values of 0x0 in the vehicle capture. The foreground and background flows entered the measured path without an observed packet-class difference between the requested labels.

D.5 Concurrent-Client Results

In the –106-dBm seven-point repeat, 100 clients produced aggregate p95 RTTs of 81.850 ms for MQTT, 264.070 ms for TCP, and 113.391 ms for UDP. MQTT and TCP each returned 100,000 of 100,000 responses. UDP returned 99,965 of 100,000. A separate 100-client repeat yielded p95 values of 177.053, 283.765, and 145.894 ms and returned 100,000, 100,000, and 99,365 responses.

The 100-client collection at –120 dBm produced a different ordering. MQTT and TCP p95 rose to 5,135.984 and 2,908.420 ms. They completed 99.499% and 100% of issued requests before timeout. UDP completed 90,811 of 100,000 requests, with a p95 of 172.500 ms. Its shorter response tail coincides with 9,189 missing cycles.

The per-client distributions reveal substantial unfairness within the shared attachment. Across 100 clients, the median client p95/p99 was 3,814.102/13,149.421 ms for MQTT and 2,938.408/6,400.146 ms for TCP. Many flows fall outside the 100- and 200-ms application budgets even though most requests still receive a response before timeout.